

Cauchy's Integral Formula, Revised.

Thm. Let $f: G \rightarrow \mathbb{C}$ be a Holomorphic on open $G \subset \mathbb{C}$.

If $z_0 \in G$ and $\overline{B_r(z_0)} \subseteq G$, then for a Curve $C: z_0 + re^{it}$ ($0 \leq t \leq 2\pi$),

$$f(z_0) = \frac{1}{2\pi i} \cdot \int_C \frac{f(z)}{z - z_0} dz.$$

Proof. Denote $D = B_r(z_0)$. Let $z_0 \in D$ be fixed, and consider a Curve $\Gamma_{\delta, \epsilon}$ as

Since $\frac{f(z)}{z - z_0}$ is Holomorphic on $D \setminus \{z_0\}$, $\int_{\Gamma_{\delta, \epsilon}} \frac{f(z)}{z - z_0} dz = 0$.



Taking $\delta \rightarrow 0$, then we obtain

$$\lim_{\delta \rightarrow 0} \int_{\Gamma_{\delta, \epsilon}} \frac{f(z)}{z - z_0} dz = - \int_{C_\epsilon} \frac{f(z)}{z - z_0} dz + \int_C \frac{f(z)}{z - z_0} dz = 0.$$

Meanwhile, $\frac{f(z)}{z - z_0} = \frac{f(z) - f(z_0)}{z - z_0} + \frac{f(z_0)}{z - z_0}$ and since f is Holomorphic,

the first term is bounded, therefore $\lim_{\epsilon \rightarrow 0} \int_{C_\epsilon} \frac{f(z) - f(z_0)}{z - z_0} dz = 0$.

$$\text{Now, } \lim_{\epsilon \rightarrow 0} \int_{C_\epsilon} \frac{f(z_0)}{z - z_0} dz = f(z_0) \cdot \lim_{\epsilon \rightarrow 0} \int_{C_\epsilon} \frac{1}{z - z_0} dz$$

$$= f(z_0) \cdot \lim_{\epsilon \rightarrow 0} \left(\int_0^{2\pi} \frac{1}{\epsilon \cdot e^{i\epsilon}} \cdot \epsilon \cdot i \cdot e^{i\epsilon} dt \right)$$

$$= f(z_0) \cdot 2\pi i.$$

$$\int_C \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0).$$

Consequently,

$$\int_C \frac{f(z)}{z - z_0} dz = \lim_{\delta \rightarrow 0} \int_{\Gamma_{\delta, \epsilon}} \frac{f(z)}{z - z_0} dz + \int_{C_\epsilon} \frac{f(z)}{z - z_0} dz = 2\pi i \cdot f(z_0).$$

Thm. Suppose f is Holomorphic on an open G containing a closed disk $\overline{D}$ having center z_0 ,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n \quad (z \in D)$$

Proof.
$$f(z) = \frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{\mu - z} d\mu \quad \left(\frac{1}{\mu - z} = \frac{1}{\mu - z_0 - (z - z_0)} = \frac{1}{\mu - z_0} \cdot \frac{1}{1 - \left(\frac{z - z_0}{\mu - z_0}\right)} \right)$$

$$= \frac{1}{2\pi i} \cdot \int_C f(\mu) \cdot \sum_{n=0}^{\infty} \frac{1}{(\mu - z_0)^{n+1}} \cdot (z - z_0)^n d\mu$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{(\mu - z_0)^{n+1}} d\mu \right) \cdot (z - z_0)^n$$

$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n$$

Conclary, Liouville's Theorem.

If $f: \mathbb{C} \rightarrow \mathbb{C}$ is entire and bounded, then it is constant.

Proof. Given $z_0 \in \mathbb{C}$ and $R > 0$, $|f'(z_0)| \leq \frac{1}{R} \cdot \sup_{C_R} |f| \rightarrow 0$ as $R \rightarrow \infty$. Therefore, $f' = 0$.
 Thus f is constant.

Fundamental Theorem of Algebra.

Non-constant polynomial $P(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_0 \in \mathbb{C}[z]$ has a root in $\mathbb{C}$.

Proof. Suppose that $P(z)$ has no root in $\mathbb{C}$. Then, $1/P(z)$ is well-defined on $\mathbb{C}$, entire. observe that
$$\frac{P(z)}{z^n} = a_n + \left(\frac{a_{n-1}}{z} + \dots + \frac{a_0}{z^n} \right) \quad (z \neq 0).$$

The terms $a_{n-k} \cdot z^{-k}$ ($k=1, \dots, n$) converges to zero as $|z| \rightarrow \infty$, thus for some $R > 0$,

$$\begin{aligned} |P(z)| &\geq 2|a_n| \cdot z^n \quad (|z| > R) \\ &\geq 2|a_n| \cdot R^n \end{aligned}$$

Therefore it is bounded below in $|z| > R$, and clearly in $|z| \leq R$.

Now, $\frac{1}{P}$ is bounded, thus constant. Contradiction.